KAIZEN MATHS

Lesson starter

Momentum and Impulse

Collisions and Restitution

Name

S1

A particle of mass 3 kg has velocity 5 m s^{-1} to the right before a collision and 3 m s^{-1} to the right after the collision. Find the impulse exerted on the particle.

S2

Particle A moves to the right at 8 m s^{-1} and particle B moves to the right at 3 m s^{-1}. After impact, A has velocity 0 m s^{-1} and B has velocity 3.75 m s^{-1} to the right. Find the coefficient of restitution.

S3

A ball of mass 0.1 kg strikes a fixed vertical wall at 8 m s^{-1} perpendicular to the wall. The coefficient of restitution is 0.8. Find the impulse exerted on the ball.

S4

Two particles A and B of masses 4 kg and 5 kg move in the same straight line. Before collision, their velocities are 6 m s^{-1} and 2 m s^{-1} to the right. The coefficient of restitution is 0.75. Find their velocities after collision.

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KAIZEN MATHS

Exit ticket

Momentum and Impulse

Collisions and Restitution

Name _____

EA1

A particle of mass 0.5 kg has velocity 10 m s^{-1} to the right before a collision and 3 m s^{-1} to the right after the collision. Find the impulse exerted on the particle.

EA2

Particle A moves to the right at 12 m s^{-1} and particle B moves to the right at 2 m s^{-1}. After impact, A has velocity 1 m s^{-1} to the right and B has velocity 3 m s^{-1} to the right. Find the coefficient of restitution.

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Exit ticket

Momentum and Impulse

Collisions and Restitution

Name _____

EB1

A ball of mass 0.2 kg strikes a fixed vertical wall at 12 m s^{-1} perpendicular to the wall. The coefficient of restitution is 0.8. Find the impulse exerted on the ball.

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Momentum and Impulse

Collisions and Restitution

Name _____

EA1

A particle of mass 0.5 kg has velocity 10 m s^{-1} to the right before a collision and 3 m s^{-1} to the right after the collision. Find the impulse exerted on the particle.

EA2

Particle A moves to the right at 12 m s^{-1} and particle B moves to the right at 2 m s^{-1}. After impact, A has velocity 1 m s^{-1} to the right and B has velocity 3 m s^{-1} to the right. Find the coefficient of restitution.

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Momentum and Impulse

Collisions and Restitution

Name _____

EB1

A ball of mass 0.2 kg strikes a fixed vertical wall at 12 m s^{-1} perpendicular to the wall. The coefficient of restitution is 0.8. Find the impulse exerted on the ball.

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Teacher answers & working

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S1

A particle of mass 3 kg has velocity 5 m s^{-1} to the right before a collision and 3 m s^{-1} to the right after the collision. Find the impulse exerted on the particle.

Answer:

6 N s to the left

Working:

Take right as positive.

$u = 5, v = 3, m = 3$

Impulse on a particle equals final momentum minus initial momentum.

Impulse = mv - mu

Impulse = 3(3) - 3(5)

Impulse = 9 - 15

Impulse = -6 N s

Impulse = 6 N s to the left.

S2

Particle A moves to the right at 8 m s^{-1} and particle B moves to the right at 3 m s^{-1}. After impact, A has velocity 0 m s^{-1} and B has velocity 3.75 m s^{-1} to the right. Find the coefficient of restitution.

Answer:

$e = 0.75$

Working:

Coefficient of restitution is speed of separation divided by speed of approach.

e = speed of separation ÷ speed of approach

Before impact, A is catching B, so the approach speed is $u_1 - u_2$.

speed of approach = 8 - 3 = 5 m s^{-1}

After impact, B is moving away from A, so the separation speed is $v_2 - v_1$.

speed of separation = 3.75 - (0) = 3.75 m s^{-1}

$e = 3.75 \div 5$

$e = 0.75$

The coefficient of restitution is 0.75.

Teacher answers & working

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S3

A ball of mass 0.1 kg strikes a fixed vertical wall at 8 m s^{-1} perpendicular to the wall. The coefficient of restitution is 0.8. Find the impulse exerted on the ball.

Answer:

1.44 N s away from the wall

Working:

Take motion towards the wall as positive.

For a fixed wall, the rebound speed is e times the approach speed in the opposite direction.

$v = -eu = -0.8(8) = -6.4$ m s^{-1}

Impulse equals final momentum minus initial momentum.

Impulse = m(v - u)

Impulse = 0.1(-6.4 - 8)

Impulse = -1.44 N s

The negative sign means the impulse acts away from the wall.

Impulse = 1.44 N s away from the wall.

Teacher answers & working

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S4

Two particles A and B of masses 4 kg and 5 kg move in the same straight line. Before collision, their velocities are 6 m s^{-1} and 2 m s^{-1} to the right. The coefficient of restitution is 0.75. Find their velocities after collision.

Answer:

A: 2.111 m s^{-1} to the right, B: 5.111 m s^{-1} to the right

Working:

Take right as positive. Let the final velocities of A and B be v_1 and v_2.

Use conservation of momentum for the first equation.

$4(6) + 5(2) = 4v_1 + 5v_2$

$34 = 4v_1 + 5v_2$

Use restitution for the second equation.

e = speed of separation ÷ speed of approach

$v_2 - v_1 = 0.75(6 - 2)$

$v_2 - v_1 = 3$

Substitute $v_2 = v_1 + 3$ into the momentum equation.

$34 = 4v_1 + 5(v_1 + 3)$

$34 = 9v_1 + 15$

$9v_1 = 19$

$v_1 = 2.111 \text{ m s}^{-1}$

$v_2 = 2.111 + 3 = 5.111 \text{ m s}^{-1}$

A has velocity 2.111 m s^{-1} to the right and B has velocity 5.111 m s^{-1} to the right.

EA1

A particle of mass 0.5 kg has velocity 10 m s^{-1} to the right before a collision and 3 m s^{-1} to the right after the collision. Find the impulse exerted on the particle.

Answer:

3.5 N s to the left

Working:

Take right as positive.

$u = 10, v = 3, m = 0.5$

Impulse on a particle equals final momentum minus initial momentum.

Impulse = mv - mu

Impulse = 0.5(3) - 0.5(10)

Impulse = 1.5 - 5

Impulse = -3.5 N s

Impulse = 3.5 N s to the left.

Teacher answers & working

Momentum and Impulse · Collisions and Restitution

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EA2

Particle A moves to the right at 12 m s^{-1} and particle B moves to the right at 2 m s^{-1}. After impact, A has velocity 1 m s^{-1} to the right and B has velocity 3 m s^{-1} to the right. Find the coefficient of restitution.

Answer:

$e = 0.2$

Working:

Coefficient of restitution is speed of separation divided by speed of approach.

e = speed of separation ÷ speed of approach

Before impact, A is catching B, so the approach speed is $u_1 - u_2$.

speed of approach = 12 - 2 = 10 m s^{-1}

After impact, B is moving away from A, so the separation speed is $v_2 - v_1$.

speed of separation = 3 - (1) = 2 m s^{-1}

$e = 2 \div 10$

$e = 0.2$

The coefficient of restitution is 0.2.

EB1

A ball of mass 0.2 kg strikes a fixed vertical wall at 12 m s^{-1} perpendicular to the wall. The coefficient of restitution is 0.8. Find the impulse exerted on the ball.

Answer:

4.32 N s away from the wall

Working:

Take motion towards the wall as positive.

For a fixed wall, the rebound speed is e times the approach speed in the opposite direction.

$v = -eu = -0.8(12) = -9.6$ m s^{-1}

Impulse equals final momentum minus initial momentum.

Impulse = m(v - u)

Impulse = 0.2(-9.6 - 12)

Impulse = -4.32 N s

The negative sign means the impulse acts away from the wall.

Impulse = 4.32 N s away from the wall.